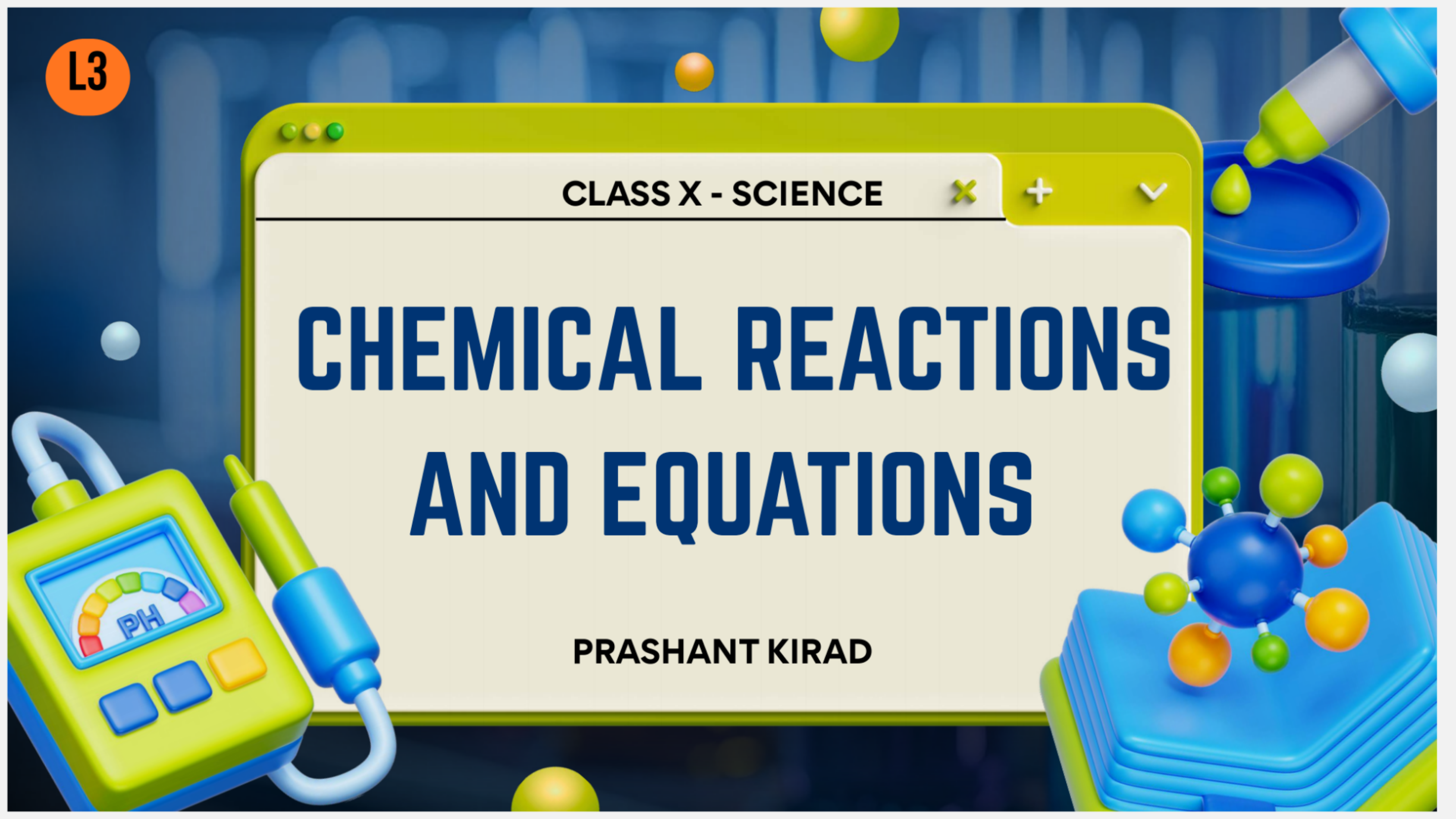


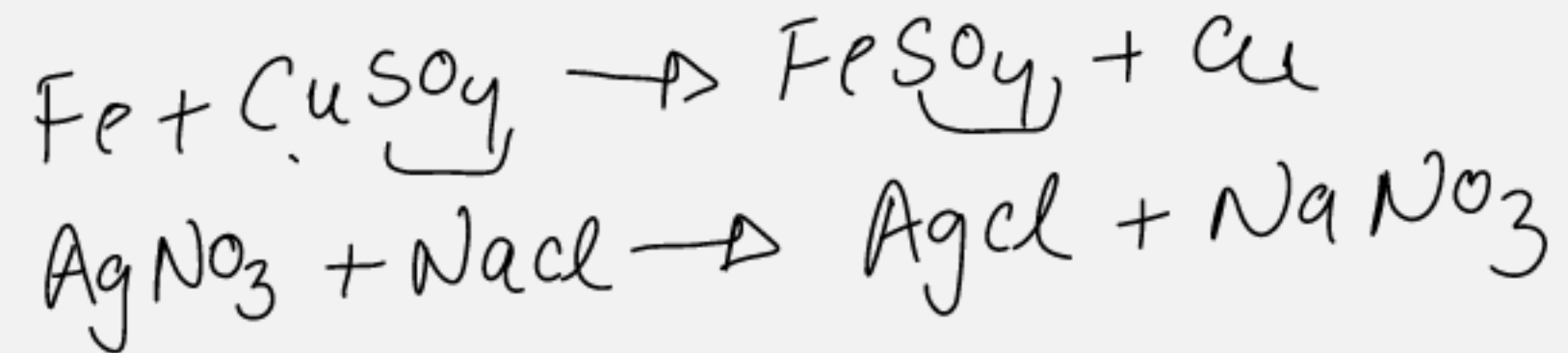
L3

CLASS X - SCIENCE

CHEMICAL REACTIONS AND EQUATIONS

PRASHANT KIRAD





HOMEWORK

आरंभ
BATCH

(a) Hydrogen + Chlorine $\rightarrow$ Hydrogen chloride

~~(b) Barium chloride + Sodium sulphate $\rightarrow$ Barium sulphate + Sodium chloride~~

(c) Iron + Copper sulphate $\rightarrow$ Iron sulphate + Copper $\text{BaCl}_2 + \text{Na}_2\text{SO}_4$

~~(d) Propane + Oxygen $\rightarrow$ Carbon dioxide + Water $\rightarrow \text{BaSO}_4 + 2\text{NaCl}$~~

(e) Silver nitrate + Sodium chloride $\rightarrow$ Silver chloride + Sodium nitrate

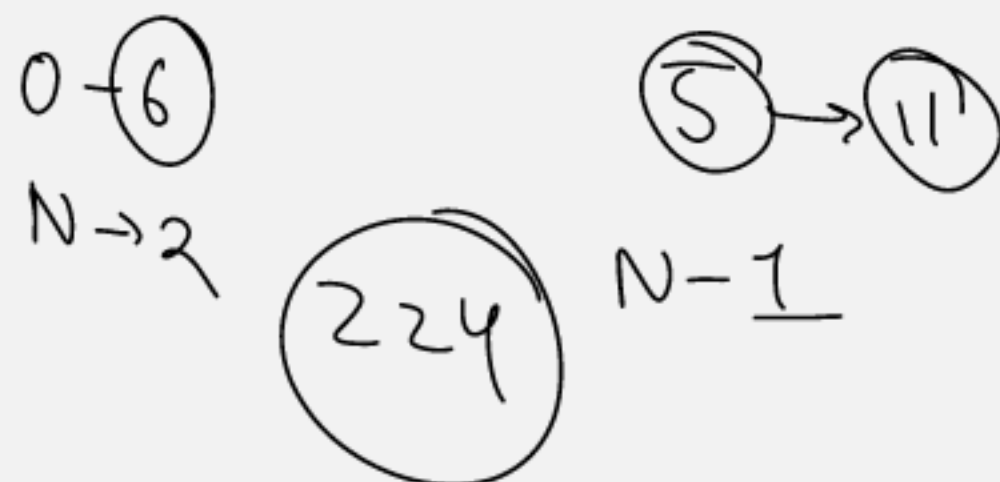
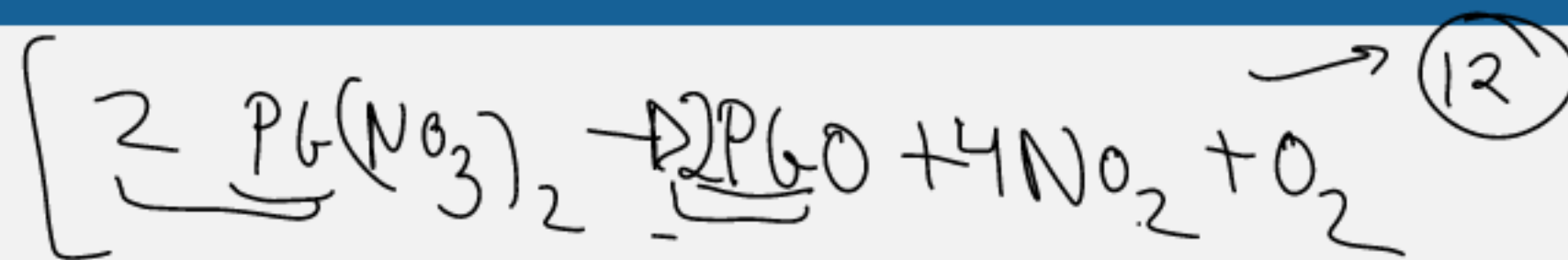
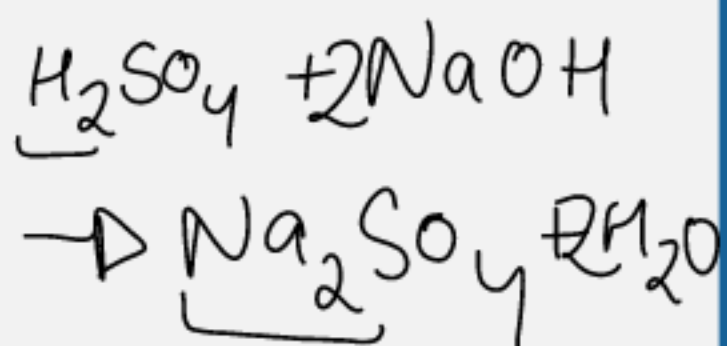
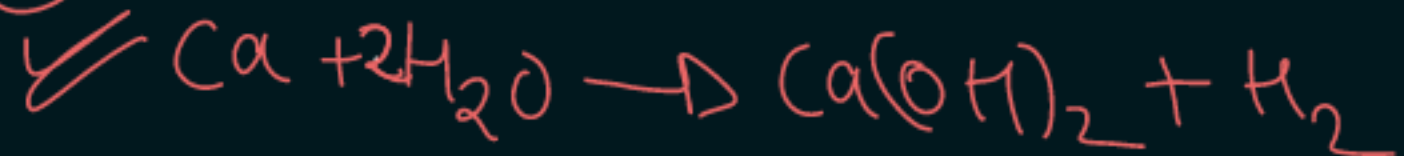
²⁰²⁷
(f) Lead nitrate $\rightarrow$ Lead oxide + Nitrogen dioxide + Oxygen

(g) Ammonia + Oxygen $\rightarrow$ Nitric oxide + Water

(h) Sulphuric acid + Sodium hydroxide $\rightarrow$ Sodium sulphate + Water

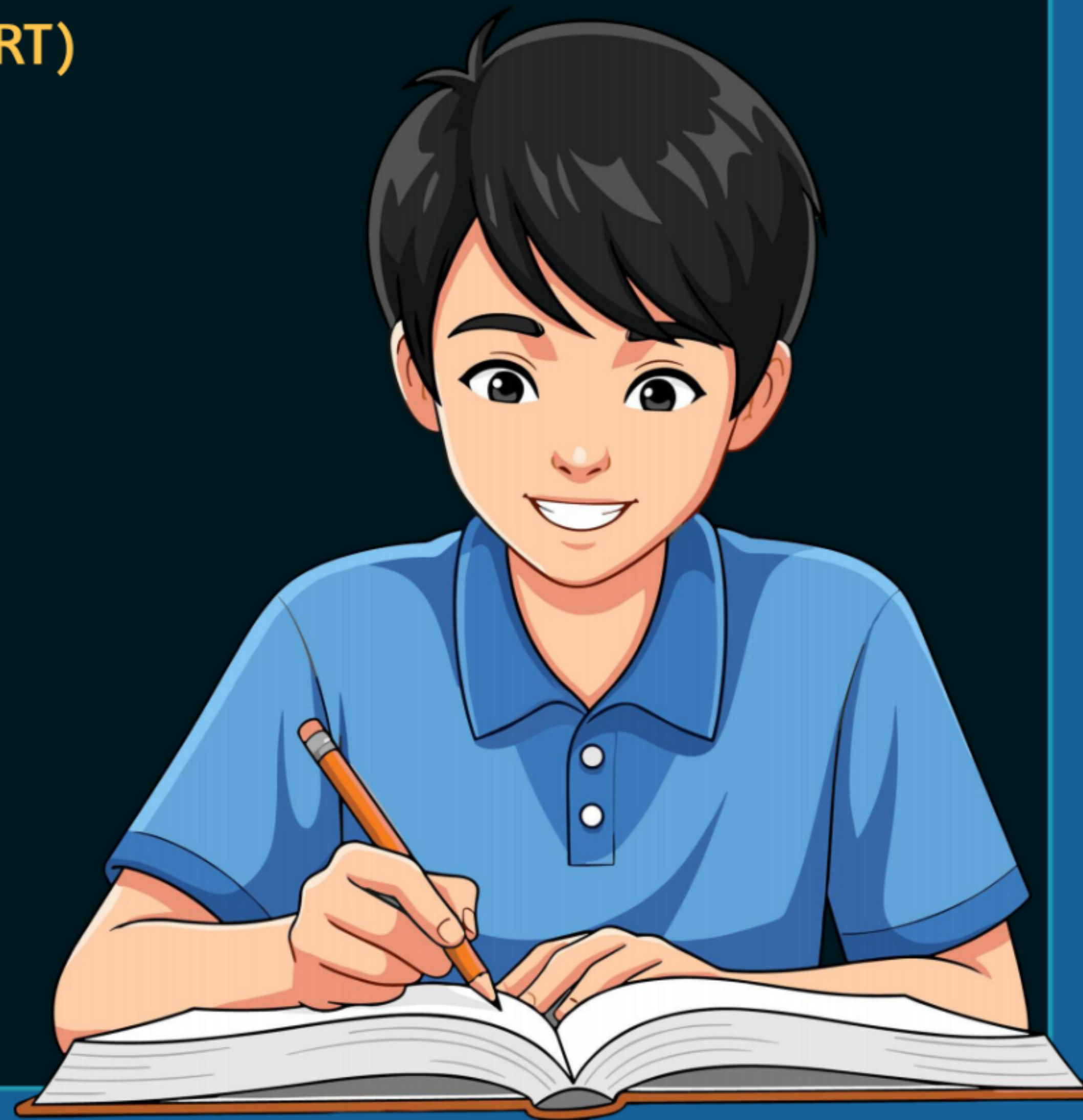
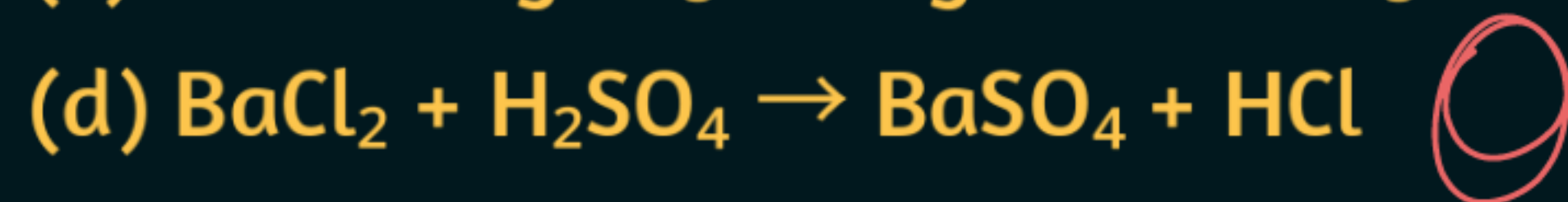
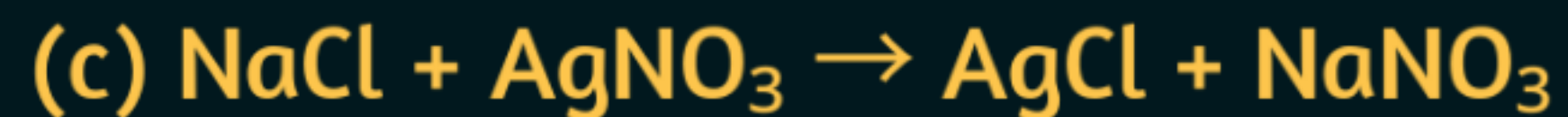
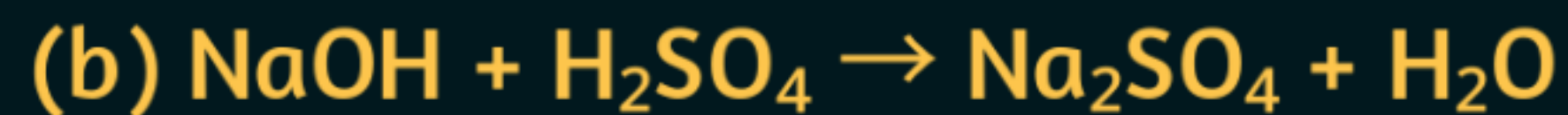
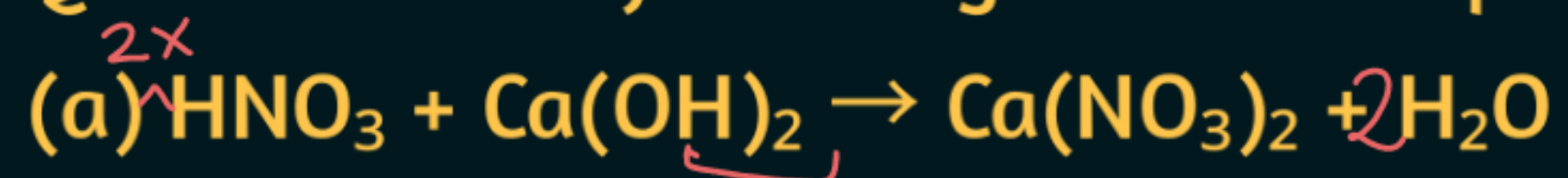
(i) Calcium + Water $\rightarrow$ Calcium hydroxide + Hydrogen

(j) Carbon + Oxygen $\rightarrow$ Carbon monoxide $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$



HOMEWORK

Q. Balance the following chemical equations. (NCERT)



LET'S TEST YOUR MEMORY



आरंभ
BATCH

Q. $\text{MnO}_2 + x\text{HCl} \rightarrow \text{MnCl}_2 + y\text{H}_2\text{O} + z\text{Cl}_2$. In order to balance the above chemical equation, the value of x, y and z respectively are :
(CBSE 2023)

A

6,2,2

B

4,2,1

C

2,2,1

D

4,1,2

LET'S TEST YOUR MEMORY



Q. Identify 'p', 'q' and 'r' in the following balanced reaction:



(CBSE 2023, 2025)

224

A

2, 2, 4

B

2, 4, 2

C

2, 2, 1

D

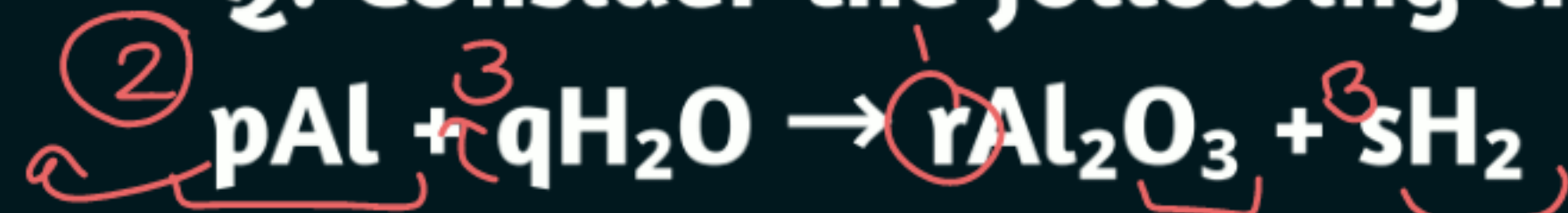
4, 1, 2

LET'S TEST YOUR MEMORY



आरंभ
BATCH

Q. Consider the following chemical equation:



To balance this equation, the values of 'p', 'q', 'r', and 's' are:

A

3, 2, 2, 1

B

2, 3, 3, 1

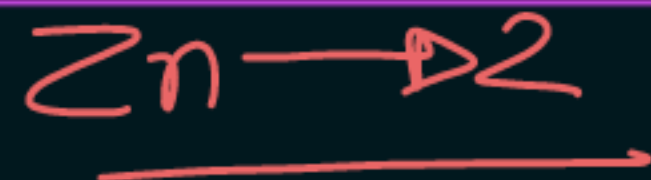
C

2, 3, 1, 3

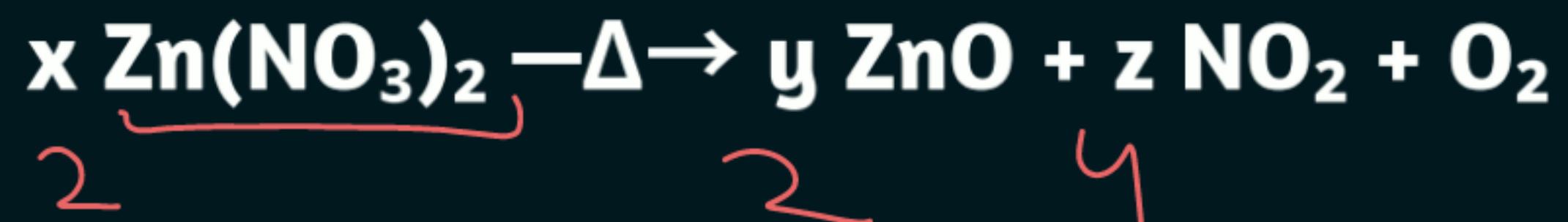
D

3, 1, 2, 2

LET'S TEST YOUR MEMORY



Q. To balance the following chemical equation, the values of the coefficients x, y and z must be respectively:



A

4, 2, 2

B

4, 4, 2

C

2, 2, 4

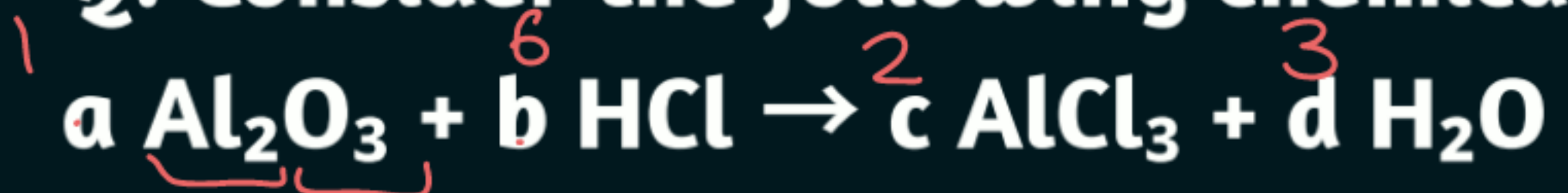
D

2, 4, 2

LET'S TEST YOUR MEMORY



Q. Consider the following chemical equation: (2024)



In order to balance this chemical equation, the values of a, b, c and d must be:

A

1, 6, 2 and 3

B

1, 6, 3 and 2

C

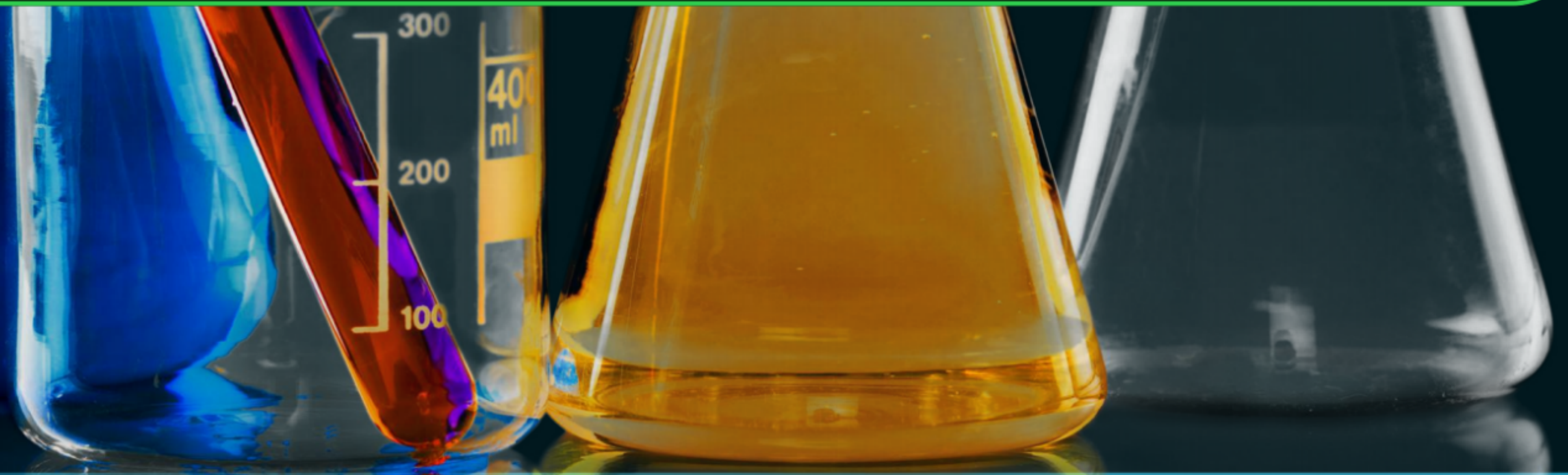
2, 6, 2 and 3

D

2, 6, 3 and 2



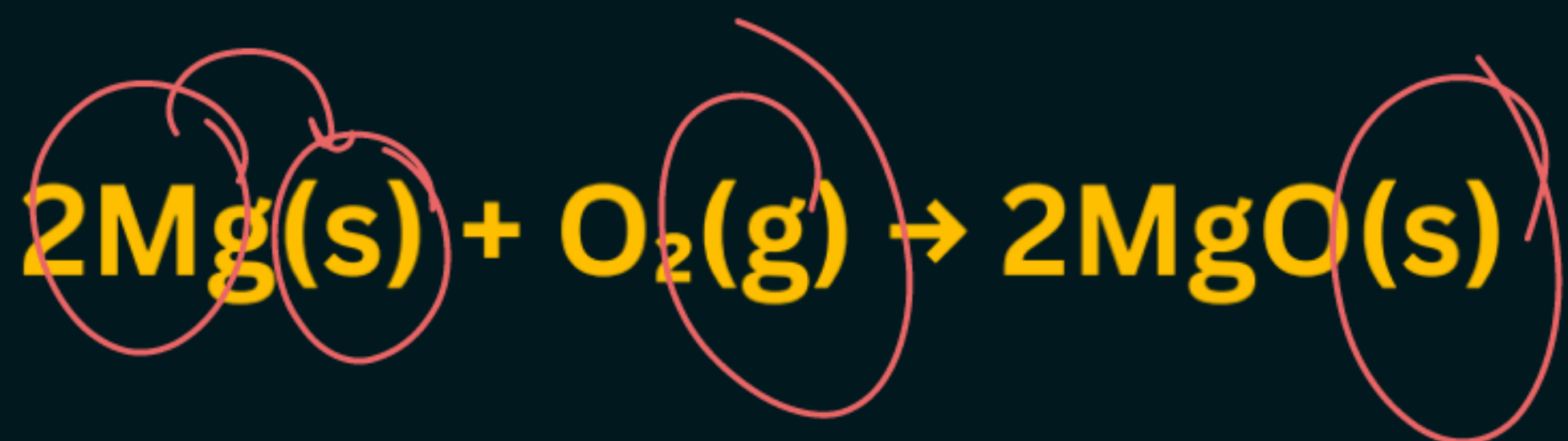
Limitations of Chemical Equations and their removal



~~(i)~~ No Physical States Mentioned:

It does not tell whether substances are solid, liquid, gas, or aqueous.

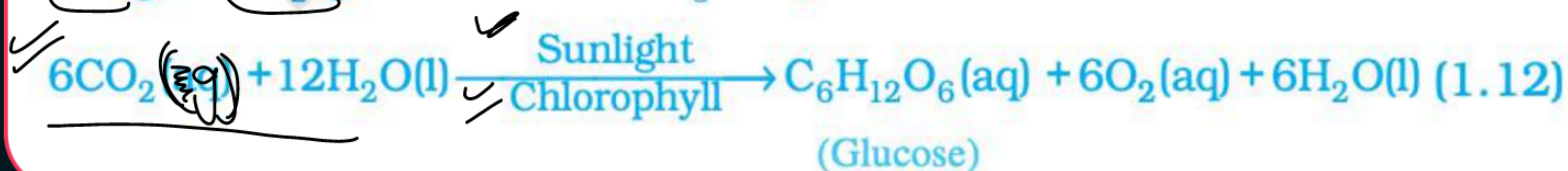
Physical State	State Symbol
Solid	s
Liquid	l
Gas	g
Aqueous Solution	aq



(ii) No Reaction Conditions

It does not indicate the conditions under which the reaction takes place, such as temperature, pressure, or catalyst.

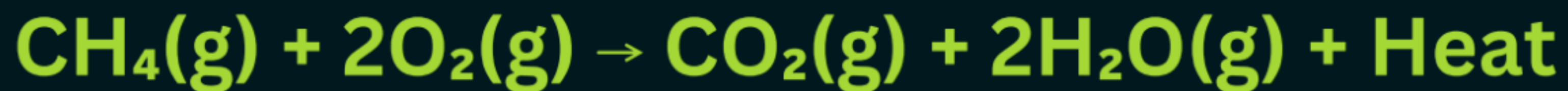
Sometimes the reaction conditions, such as temperature, pressure, catalyst, etc., for the reaction are indicated above and/or below the arrow in the equation.



(iii) No Information About Energy Change

It does not tell whether the reaction is exothermic (heat released) or endothermic (heat absorbed).

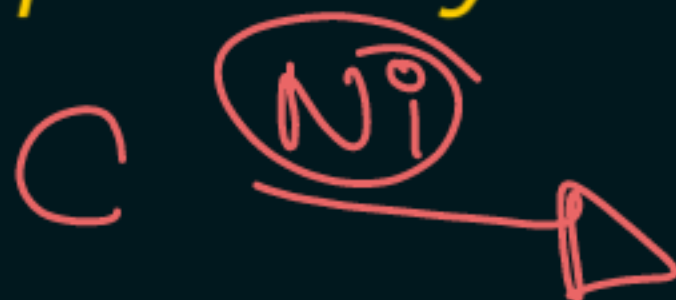
The energy change is shown by writing “+ Heat” or “+ Energy” in the equation.



(iv) No Information About Speed of Reaction

It does not indicate whether the reaction is fast or slow.

The rate of reaction is studied separately and sometimes mentioned in description.



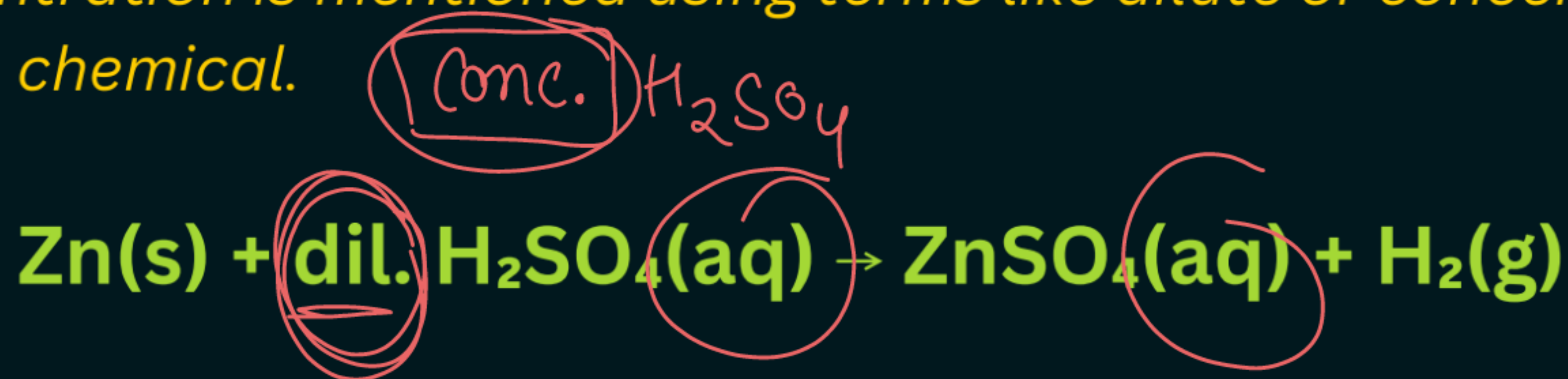
Rusting of iron is a slow reaction, while burning of magnesium is fast.



(v) No Information About Concentration of Solutions

It does not specify whether the solution is dilute or concentrated.

The concentration is mentioned using terms like dilute or concentrated before the chemical.



TYPES OF CHEMICAL REACTIONS

Combination
Reaction

Redox
Reaction

Decomposition
Reaction

Displacement
Reaction

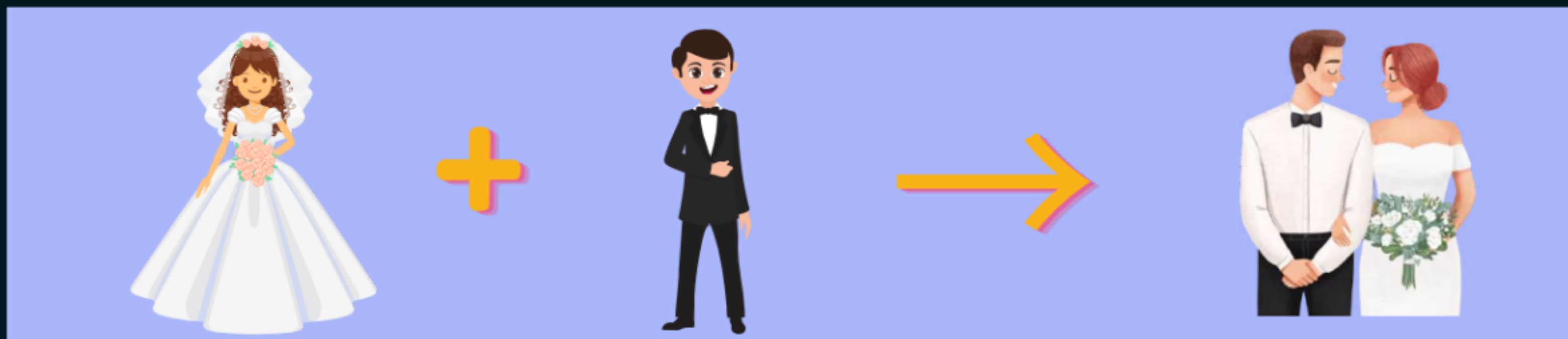
Double-
Displacement
Reaction



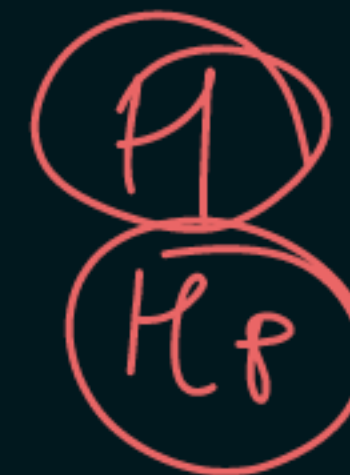
2027
⊗ ⊗

Combination Reaction

A combination or synthesis reaction is a chemical reaction where two or more substances combine to form a single new substance.



Combination Reaction



Types of combination reactions possible:

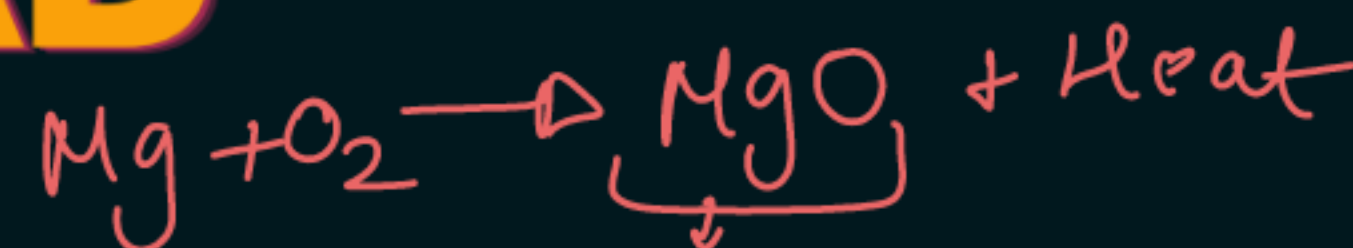
- Combination of two elements to form a compound.
- Combination of two compounds to form a compound.
- Combination of an element and a compound to form a compound.

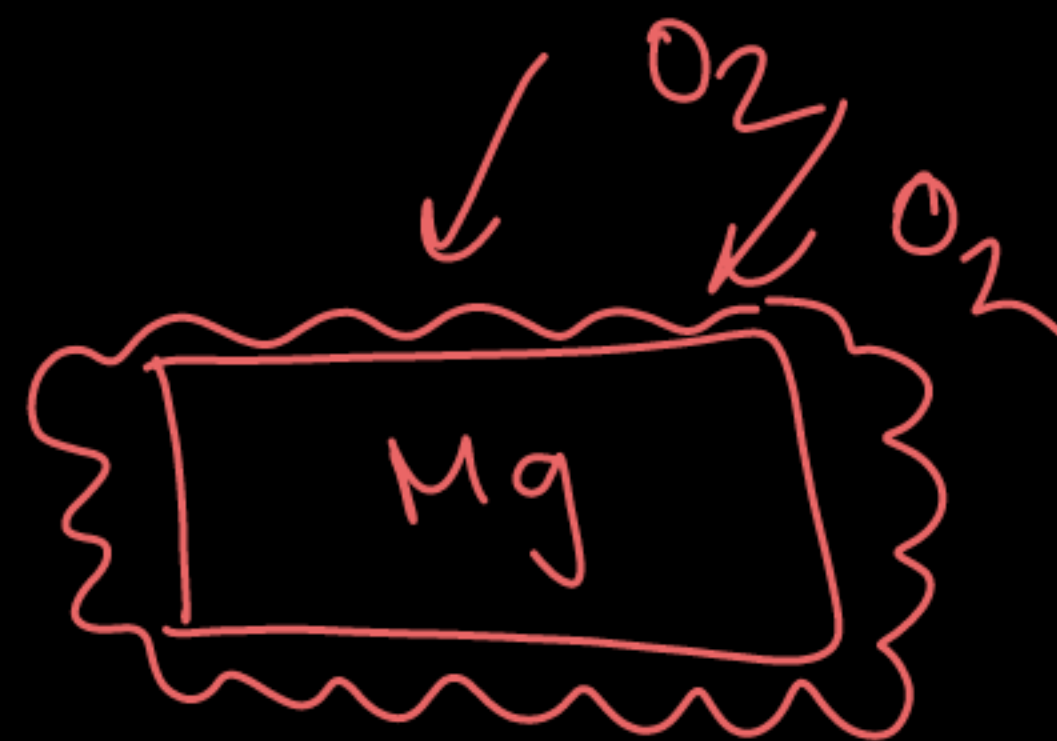
i. Combination of two elements to form a compound



Example:

[Burning of Magnesium Ribbon]





Activity 1.1

CAUTION: This Activity needs the teacher's assistance. It would be better if students wear suitable eyeglasses.

- Clean a magnesium ribbon about 3-4 cm long by rubbing it with sandpaper.
- Hold it with a pair of tongs. Burn it using a spirit lamp or burner and collect the ash so formed in a watch-glass as shown in Fig. 1.1. Burn the magnesium ribbon keeping it away as far as possible from your eyes.
- What do you observe?

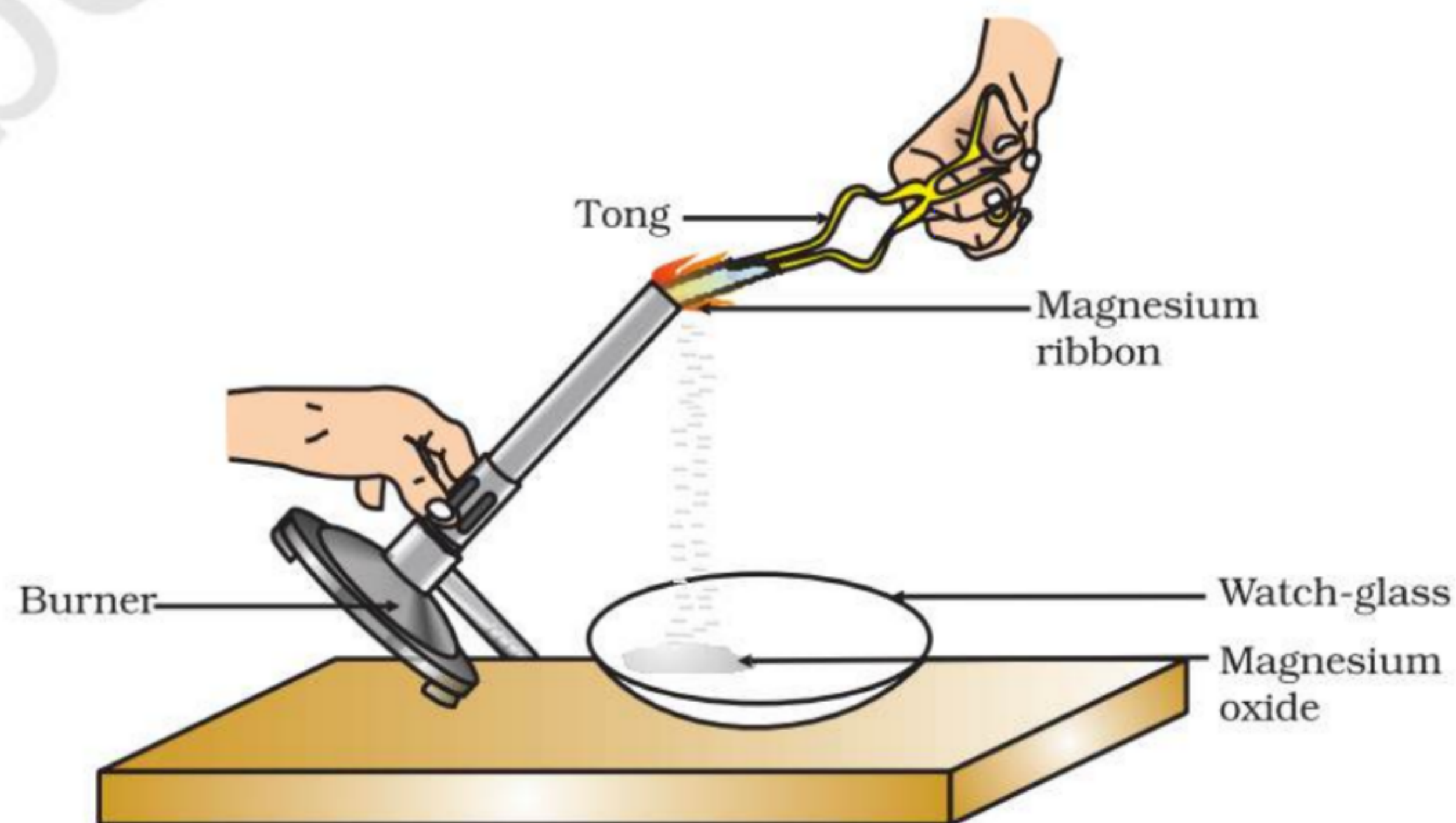


Figure 1.1

Burning of a magnesium ribbon in air and collection of magnesium oxide in a watch-glass

PK Points!



- ✓ Magnesium ribbon is cleaned with sandpaper to remove **oxide layer**.
- ✓ It burns with a dazzling **white flame**.
- ✓ **White powder** (magnesium oxide) is formed.

Reaction:

Magnesium + Oxygen $\rightarrow$ Magnesium oxide



- Type of reaction: **Combination and Exothermic**.



ii. Combination of two compounds to form a compound



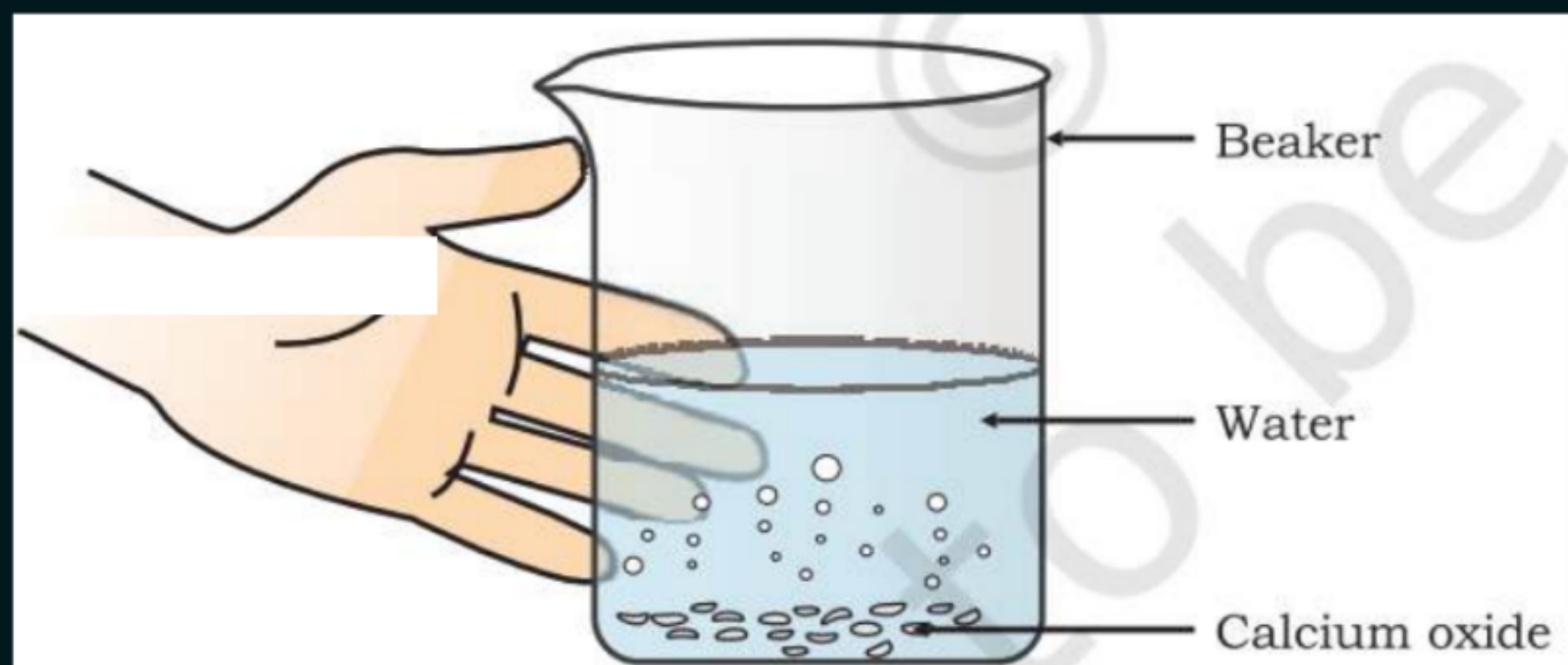
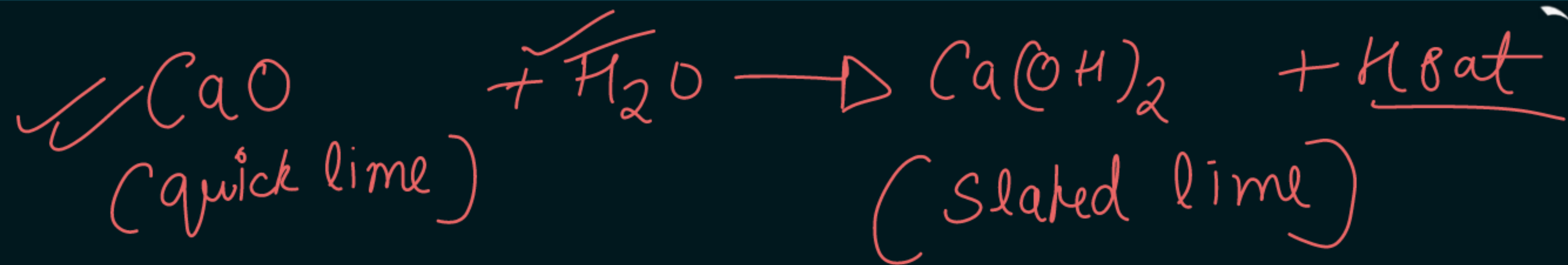


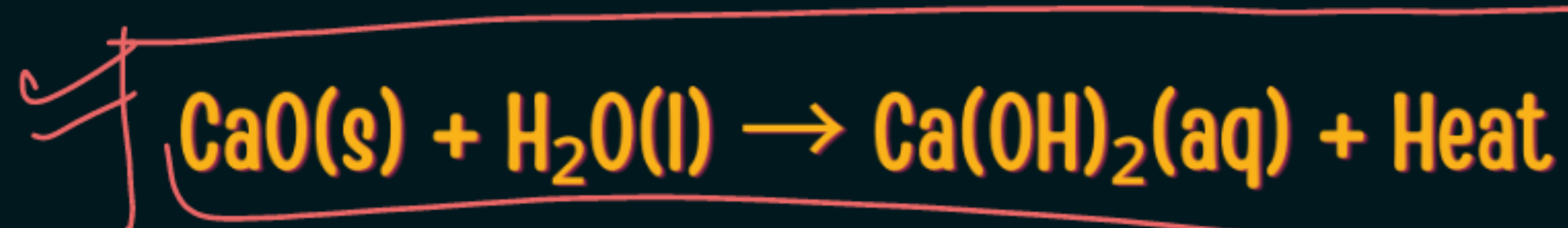
Figure 1.3

Activity 1.4

- Take a small amount of calcium oxide or quick lime in a beaker.
- Slowly add water to this.
- Touch the beaker as shown in Fig. 1.3.
- Do you feel any change in temperature?

ii. Combination of two compounds to form a compound

Example: Quicklime (CaO) + Water (H₂O)

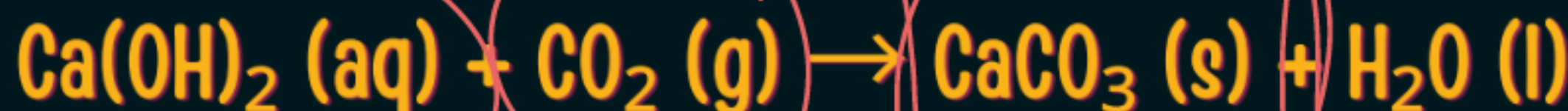


lime water

Shiny

Explanation: Quick lime reacts with water to form slaked lime (Calcium hydroxide), releasing heat (exothermic reaction).

Use in White washing: Slaked lime reacts with carbon dioxide in the air after 2-3 days, forming a shiny layer of calcium carbonate (CaCO₃) on walls:



Observations when quicklime is added to water

Example: Quicklime (CaO) + Water (H₂O) $\rightarrow$ Ca(OH)₂

- i. Very fast reaction (vigorous)
- ii. Heat is evolved (exothermic reaction)
- iii. Water boils to form steam and produces a hissing sound.

ii. Combination of two compounds to form a compound

Slaked lime reacts with carbon dioxide in the air after 2-3 days, forming a shiny layer of calcium carbonate (CaCO_3) on walls:



WOW!



Me looking at whitewash walls

White Wash



PK Points!



- CaO (Quick lime) reacts with H_2O to form Ca(OH)_2 (Slaked lime).
- Reaction: $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2 + \text{Heat}$
- ✓ • **Exothermic and Combination reaction.**
- ✓ • Heat is released, beaker feels hot.
- ✓ • Slaked lime is used for whitewashing.
- After 2–3 days, Ca(OH)_2 reacts with CO_2 to form CaCO_3 (Calcium carbonate): $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
- CaCO_3 gives a **shiny finish**; it's also the formula of marble.

NO STRESS



JUST VIBING